

SEASONAL AGRICULTURE PERFORMANCE ANALYSIS

Major Project – VOIS × AICTE

Submitted By

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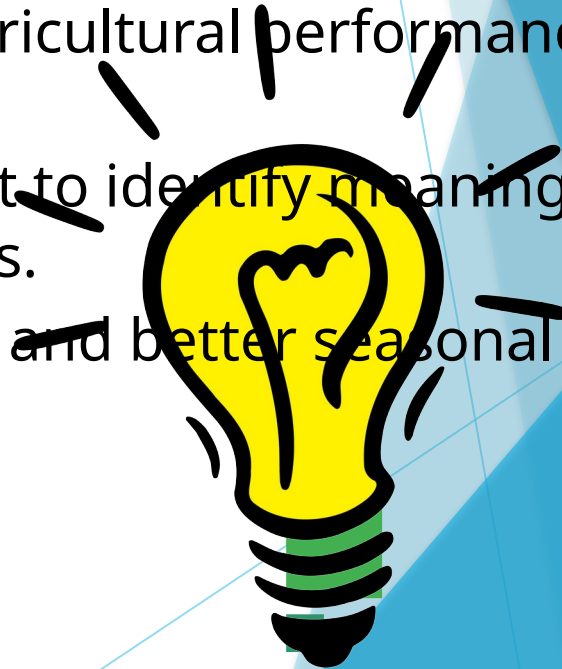
PROBLEM STATEMENT

Agricultural performance is influenced by seasonal variations in environmental conditions, farming practices, resource availability and market conditions.

Raw agricultural data does not clearly explain how agricultural performance changes from one season to another.

This project analyzes the provided agricultural dataset to identify meaningful seasonal patterns, trends, relationships and variations.

The analysis supports evidence-based understanding and better seasonal agricultural planning.





Seasonal Agriculture Performance Analysis examines 4,000 agricultural records across Kharif, Rabi and Zaid seasons. The project covers data exploration, data cleaning, descriptive analysis, seasonal comparisons, visualization and relationship analysis. Yield, production, profit, rainfall, water efficiency and disease/pest risk are studied to identify meaningful seasonal differences and trends.

WHO ARE THE END USERS?

Farmers and agricultural planners

Agricultural stakeholders and decision-makers

Data analysts and researchers

Students working with agricultural data

Users interested in evidence-based seasonal agricultural planning

Technology Used

Python

Pandas — data cleaning, transformation and aggregation

NumPy — numerical analysis

Matplotlib — data visualization

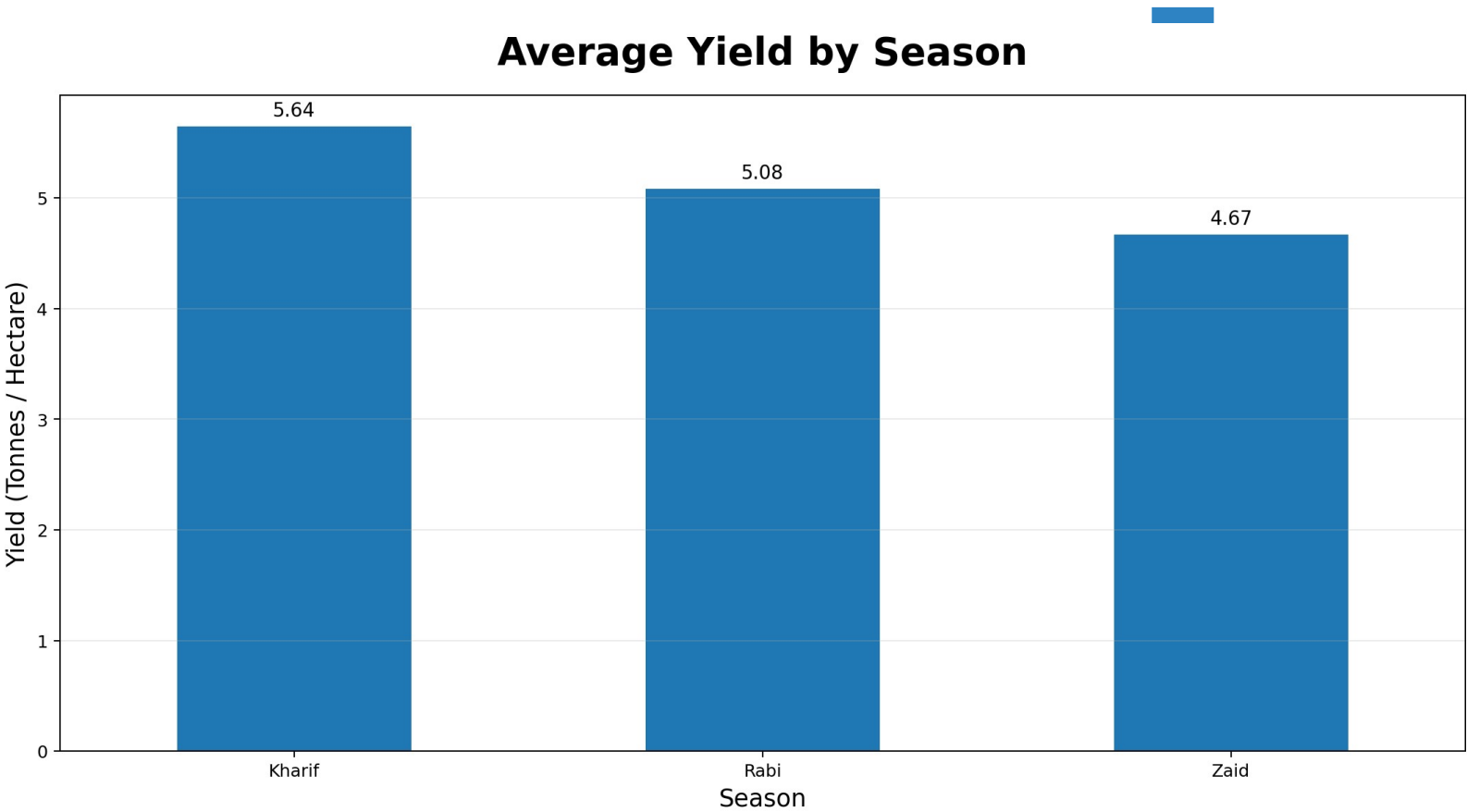
Google Colab / Jupyter Notebook — analysis and documentation

CSV — source dataset



RESULTS

RESULT 1 — Average Yield by Season

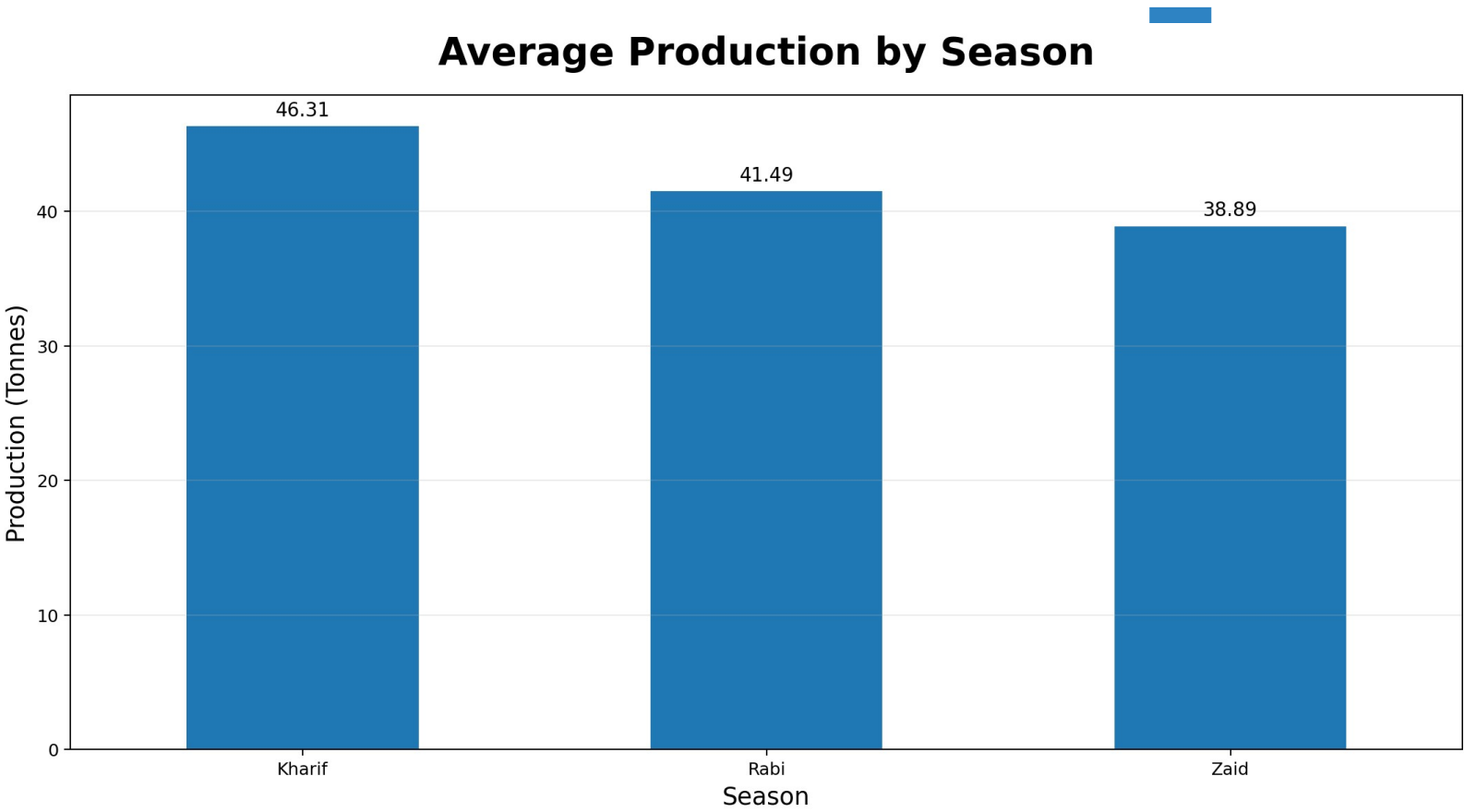


Source: VOIS Seasonal Agriculture Performance Dataset

Kharif records the highest average yield at 5.64 tonnes/hectare.

RESULTS

RESULT 2 — Average Production by Season

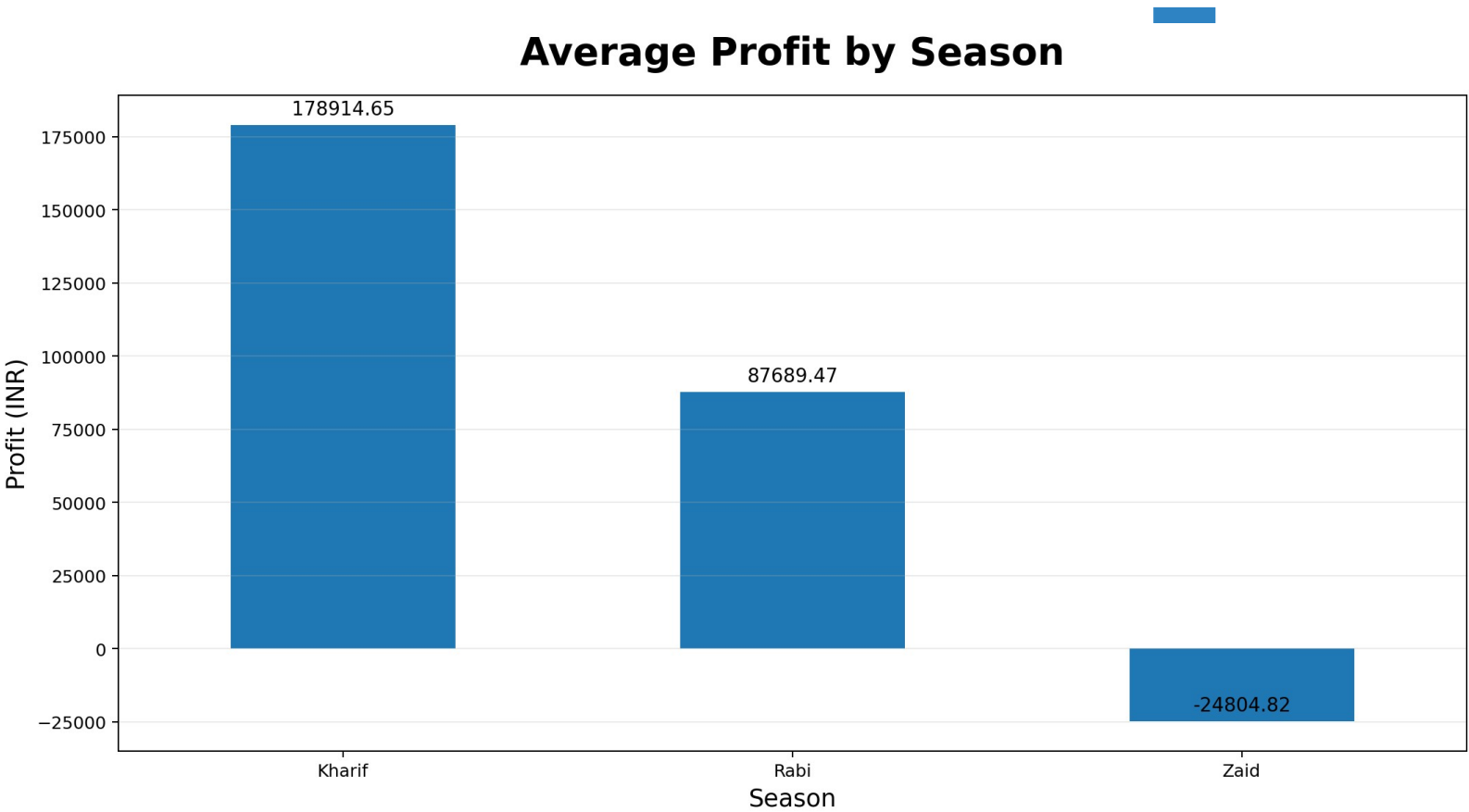


Source: VOIS Seasonal Agriculture Performance Dataset

Kharif has the highest average production at 46.31 tonnes.

RESULTS

RESULT 3 — Average Profit by Season

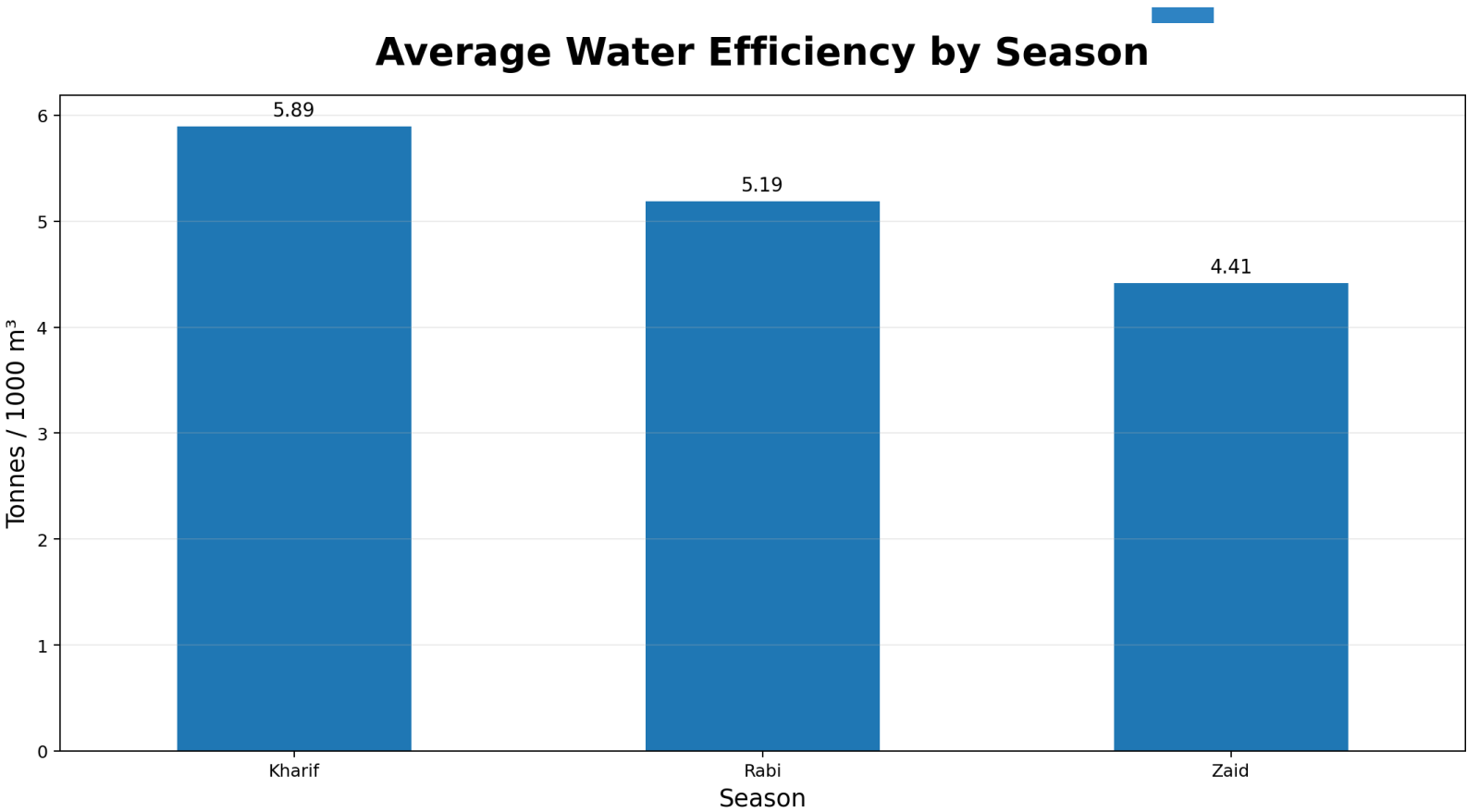


Source: VOIS Seasonal Agriculture Performance Dataset

Kharif has the highest average profit at ₹178,915.

RESULTS

RESULT 4 — Water Efficiency by Season

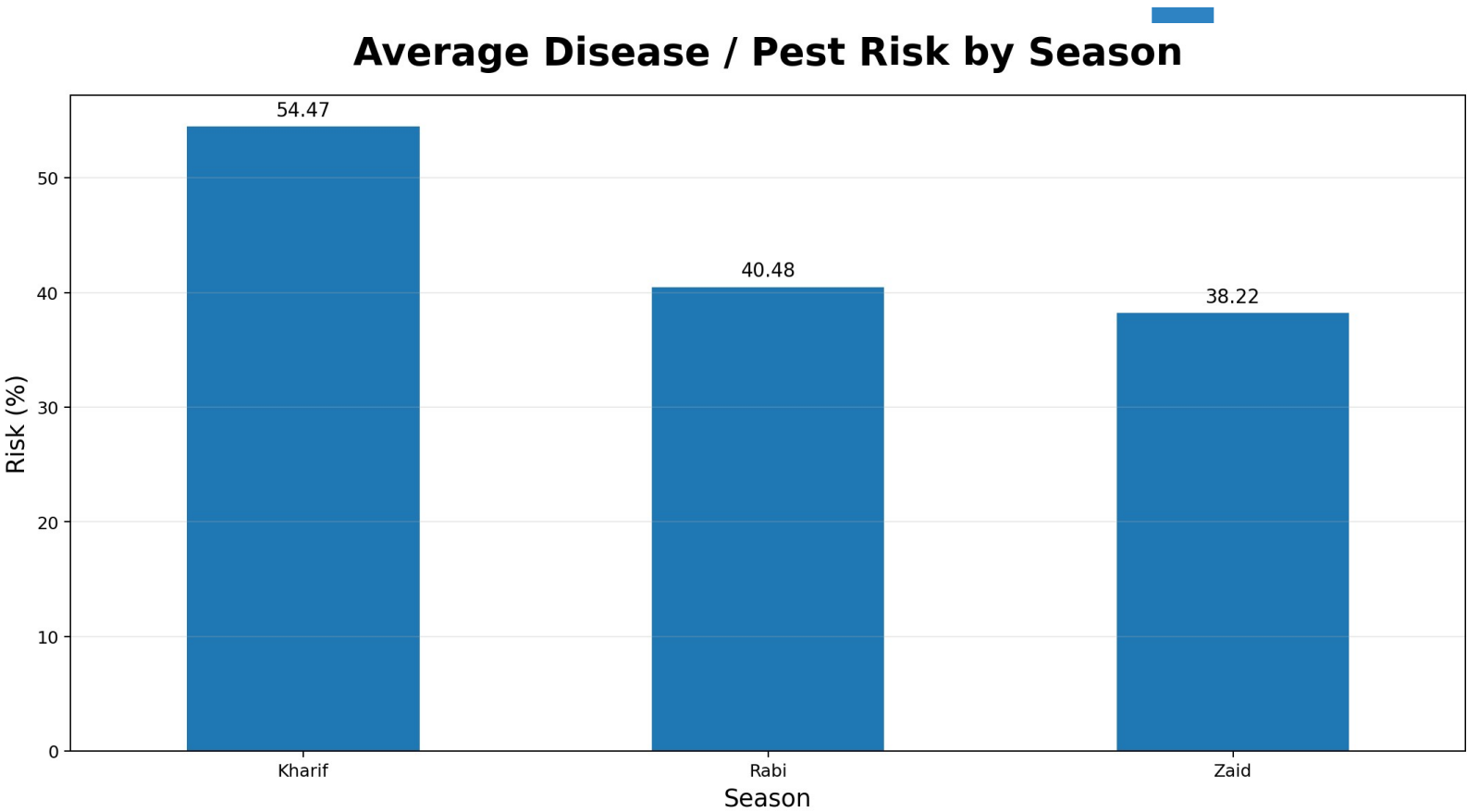


Source: VOIS Seasonal Agriculture Performance Dataset

Average water efficiency is 5.89 for Kharif, 5.19 for Rabi and 4.41 tonnes/1000 m³ for Zaid.

RESULTS

RESULT 5 — Disease / Pest Risk by Season



Source: VOIS Seasonal Agriculture Performance Dataset

Kharif shows the highest average disease/pest risk at 54.47%, compared with Rabi and Zaid.

Future scope



Extend the analysis using larger and more recent agricultural datasets.
Develop interactive dashboards for seasonal agricultural monitoring.
Integrate weather and market-price data for deeper analysis.
Apply predictive analytics for yield and economic outcome forecasting.
Perform more detailed crop-, district- and state-level seasonal analysis.



GitHub Link

GitHub Repository

<https://github.com/SanketKolhe2005/Seasonal-Agriculture-Performance-Analysis>

Repository should contain the Google Colab/Jupyter Notebook, project README, dataset information, visualizations and project documentation.

**VOIS Course completion certificate -
Course name- Data Visualization**

VOIS Course Completion Certificate — Data Visualization

INSERT YOUR DATA VISUALIZATION COURSE COMPLETION CERTIFICATE HERE



Thank You



Seasonal Agriculture Performance Analysis

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